15.1 INTRODUCTION

The term "workforce" refers to the pool of employees working in any organization or industry. They are otherwise referred to as human resources (HR) and employees. This term does not include the employer and the management, only those involved in day-to-day operational activities.

The workforce is what drives the productivity and performance of the organization, along with other vital resources such as material, money, and machinery. Thus it becomes essential for organizations to ensure that the right employees are hired at the right time with the right skill and placed in the correct position with excellent locational advantage (Khang et al., 2023a).

They are the most critical asset of any organization and are considered the most significant competitive factor for benefit. Businesses function in a dynamic environment, thus making it essential to recruit the best talents to sustain the competitive edge. Earlier organizations predominantly focused on job satisfaction as an outcome of providing permanency at workspaces with a minimal temporary and contract workforce (Babasaheb et al., 2023a).

However, due to drastic changes in employee trends and expectations, organizations need help to retain talent at workspaces. For instance, the post-pandemic phase witnessed a great resignation and significantly different expectations toward physical workspace and work-life balance (Khang et al., 2023b).

Workforce management plays a vital role in ensuring that employee performances are monitored and managed well through various practices like training, ongoing performance feedback, and provisions for rewards and recognitions.

All organizations need to realize and understand that the employees need to be satisfied, engaged, and motivated; only then will they invest themselves in the firm's affairs, thus leading to significant value addition in the form of better ideations and opportunities for the organization (Snehal et al., 2023).

Such employees display organizational citizenship and commitment toward organizational goals such as efficiency in meeting targets, better sales, and better customer service, while if employees are not performing to their best, it could significantly impact the sustainability of the organization and customer perceptions.

Another vital contribution of having positively charged employees is that they tend to build a culture of sustenance among peers, thus creating a better work environment. These such employees act as examples of good work ethics and commitment at the organization. They encourage and boost the other employees' morale, fostering a constructive and higher goal-driven work environment (Babasaheb et al., 2023b).

Employees can significantly affect the employer's public perception of the brand. They are capable of building brand equity. Employees are the people who are (B and Thimmarayappa, 2017) going to interact with outsiders on behalf of the company.

If employees are treated well and their well-being is taken care of, they are more likely to speak positively about the organization. Thus, they become brand ambassadors of the organization, otherwise known as the face of the organization. The other benefits of having satisfied employees are better profitability, performance, lower attrition, and overall organizational growth (Gupta et al., 2023).

15.1.1 Importance of Measuring Workforce Performance

There are many positive organization-related outcomes with a high workforce performance. Attaining employee job satisfaction is the primary step, which in turn helps attain other organization-related products. An organization with high performance results in higher job satisfaction.

According to Khang (2004), high performance can be achieved through flexible work hours, autonomy in work, and open communication with coworkers. Similarly, another study by Garg (2019) stated that factors such as autonomy, feedback, skill variety, task significance, and task identity contribute to high work performance and mediate the relationship between workforce performance and organizational citizenship behavior.

Most organizations aim to attain employee engagement, which is becoming a critical area of focus. Through different practices such as training, rewards, recognition, and continuous feedback, team coordination helps attain work performance and boosts employee engagement (Olateju, Kuntz & Nilakant, 2018). Turnover intention is one area of concern for most organizations. This is a negative feeling an employee has about the organization, which results in the employee quitting the organization (Dalton, Todor & Krackhardt, 1982).

Operational costs and time involved in hiring are very high, and when organizations face high attrition rates, these costs go up and affect employer branding. According to Zimmerman and Darnold (2009), high workforce performance positively impacts turnover intentions. Thus, it is essential to understand and measure workforce performance, as high work performance has many positive organizational-related outcomes.

15.1.2 Methods of Measuring Workforce Performance

Employees drive businesses to success; thus, measuring and monitoring employee performance are vital. Measuring employee performance is an ongoing process in all organizations, enabling them to understand the organization's low, medium, and high performers.

There are various techniques and methods to measure the performance of employees. Performance methods are broadly classified into traditional and modern approaches. Traditional methods include ranking, paired comparison, grading, forced distribution, forced choice, checklist, critical incident, graphic scale, essay, field review, and confidential report (Jain et al., 2023).

Modern methods include management by objectives (MBO), behaviorally anchored rating scale, and cost accounting. However, each method and the two approaches have merits and demerits. Traditional methods focus more on measuring employee traits, whereas modern methods look at employee achievement and measure the work results.

Traditional methods were non-transparent, whereby employees were not informed about their performance ratings and areas of improvement, while modern approaches are more transparent with such information.

15.2 WORKFORCE ANALYTICS AND IMPORTANCE

The process of improving and enhancing decision-making through data-based intelligence is known as workforce analytics. Workforce analytics analyze employee-related data and are used in all areas of HR functions. One such area is employee performance, where workforce analytics use different types of employee data for efficiency mapping and decision-making.

According to Khang (2024), workforce analytics help measure employee performance and its impact on business results. Unlike other performance measures, workforce analytics give an in-depth understanding of the employee's efficiency and provide better information with which organizations can improve productivity (Hota & Ghosh, 2013). They include managing and employing top talent, correlating rewards to performance, and other activities. This is the art of analyzing staff-related data using sophisticated instruments and metrics.

Instead of relying exclusively on gut instinct and intuition in the present context, workforce decisions should be supported by workforce data and analytics. Many firms lack the knowledge necessary to recognize and keep great talent. They cannot thus distinguish their value in the market.

The most valuable resource for every organization is its people. Workforce analytics, a new class of business intelligence that links human capital management to financial success, is treated as performance analytics by many top organizations.

The corporate environment is changing faster than ever in today's complex, interconnected, big-data, and disruptive world. The nature of work itself is changing due to increased talent competition and changing opportunities for workers. More information on people is available now than ever, and analysis tools are more sophisticated.

The HR function must adapt to these changes by adopting an analytical mindset and becoming more quantitative. Managers can learn about various abilities through workforce analytics and plan how to use these talents for leadership development and new commercial ventures. Through workforce analytics, businesses may learn more about employees than possible, giving them a competitive edge in enhancing operational efficiency (Jyotiranjan et al., 2013).

Many entities are already recognizing the value of analytics. Improvement is made possible by the workforce in terms of potential candidates, selection procedures, training plans, performance reviews, supervisory behavior, motivation of employees, and other areas (Levenson, 2017).

According to the economist intelligence unit PWC research from 2014 (PwC, 2014), over 89% of large organizations polled had already used big data for decision-making or intended to use it in the foreseeable future (Bhambri et al., 2022).

IBM discovered that over the past two years, there had been a 40% increase in the number of chief human resource officers (CHROs) adopting predictive data analytics to make better workforce decisions across HR functions.

HR is adopting workforce analytics for three primary reasons: the need to add value to the business; the democratization and consumerization of HR; and the evolution of work (Babasaheb et al., 2023a)

15.2.1 Contributing to Add Value to the Business

In the worldwide digital economy, businesses face the rising challenge of using the massive amounts of information they currently obtain to boost their business's and their customers' value without going too far in the direction of unlawful, unethical, or undesirable use.

In order to maintain growth, organizations pay attention, evaluate their data strategy, and implement administrative mechanisms. Organizations aim to boost performance, but it falls to HR to provide value. An analytical approach is the most effective way to accomplish this.

Workforce analytics are frequently used to demonstrate ways to enhance critical HR and people-focused activities like workforce planning, hiring, performance management, training, and employee engagement (Levenson, 2017). Managers can discover more about the various abilities of their employees through workforce analytics, and they can then plan how to use these talents for leadership development and new commercial ventures.

Workforce analytics can help forecast and enhance business outcomes like sales and profitability and the effectiveness of HR processes. Genuine workforce analytics involve evaluating organizational behavior and understanding how to link it to better business success.

15.2.2 Democratization and Consumerization

Employees, managers, and executives place greater demands on the HR function. There is more demand for individualized services and information to manage the business. Customization is becoming a fundamental expectation of the workforce from their employers. The world's best businesses should comprehend the evolving workforce's characteristics.

Businesses are also evolving to reflect the attitudes of the various age groups in their workforces. Many employees would value advice on how to enhance their working environment. Obtaining the customizing experience that the consumer seeks nowadays provides enormous benefits, since job seekers are thought of as customers (Anusha B., 2017).

Moreover, decisions about their personnel must be made by managers and executives based on timely information provided by HR executives. Managers and executives require information and suggestions regarding the problems affecting the business.

The core of this revolution is workforce analytics, since HR provides data and sophisticated algorithms created from that data to aid managers and executives in making informed decisions (Khang et al., 2022b).

15.2.3 Evolution of Work

Over the past ten years, the workplace has seen constant change. Technology for HR is evolving quickly to support workforce activities (Jyotiranjan et al., 2013). Organizations are increasingly in need of workforce analytics. In order to profit from the transformation influencing the future of work, HR professionals must adapt and build momentum in the area of workforce analytics.

15.3 HR ANALYTICS VS. WORKFORCE ANALYTICS VS. PEOPLE ANALYTICS

The approach to searching for insights into people and work data is indeed pretty recent, and every person and organization tackles it differently. Even though people and work have been studied for centuries, recent advancements in technology, access to data, and analytics have made it possible to see these subjects from a brand-new perspective.

People analytics, talent analytics, workforce analytics, and HR analytics are distinct despite being used indiscriminately. Like people and talent analytics, workforce and HR analytics can be applied alternatively. Workforce analytics and people work together, yet they have multiple interpretations (Hajimahmud et al., 2022).

15.3.1 HR Analytics

Analytics for the comprehension of HR-related procedures and methods would be considered HR analytics. It is generally carried out to upgrade or improve HR practices.

15.3.2 Workforce Analytics

Naturally, workforce analytics begin by concentrating on the work itself. Workforce analytics emphasize all facets of work, and may, but need not, involve the study of people.

Although people commonly perform work, workforce analytics go beyond the individual and include understanding work processes, planning for future work, and changing the way people work. They may even include work that is not performed at all by people, such as automation or robotic processing of work. Increasingly the talent evaluation process is heavily reliant on technology.

Data gathered through workforce analytics are used to improve and refine the assessments prospective employees and existing workers must take.

15.3.3 People Analytics

People analytics begin by putting people first. While work outcomes like productivity would be considered, other individual aspects such as prejudice, well-being, and work-life balance are also measured. These kinds of evaluations are carried out to enable better judgments about people, not usually with a focus on work, value, or HR procedures.

Under people analytics, data are accumulated on an individual from the beginning of the hiring process until the day they leave their job. The data are then applied to the organizational employees and the hiring procedure to attract strong candidates, boost the performance of employees, enhance retention, and make wiser hiring choices.

15.4 IMPLEMENTING WORKFORCE ANALYTICS

HR professionals can use one of four analytical tools to learn more about the workforce in their firm. They can collect and analyze data that depict the past or the present using descriptive analytics. Based on the recent and previous data analysis, predictive analysis aids in foreseeing future results and values.

While diagnostic analysis identifies the root cause of the issue faced in the company, the prescriptive analysis makes recommendations based on predictions and historical events.

In contrast to other operational processes, workforce analytics are crucial in today's competitive environment. In the past, HR management lacked the science and analytical tools to quantify the varied prospects. Even though technology has advanced lately, engaging in workforce analytics has gained importance for several reasons.

Leaders can gain visibility into processes and results, be prepared for the majority of outcomes, determine whether any change is sustainable for their firm, and more with the help of workforce analytics.

15.5 CHALLENGES IN WORKFORCE ANALYTICS

Today, data and big data have transformed the approach to doing business. HR analytics, as a branch of analytics, are experiencing significant breakthroughs. Organizations know that data and analytics are the best approaches to effectively accomplish their business goals and objectives (Rani et al., 2023).

Due to the multiple obstacles to HR analytics implementation, there is still a great deal of uncertainty and doubt regarding the impact of these tools and models (Dahlbom et al., 2019). Because technology, organization, and people are the three primary aspects influencing HR Analytics adoption.

When these three factors are considered together, they represent the quality of the data and IT infrastructure (technology), the organization's understanding of the impact of analytics, and the skills and capabilities (people), all of which are still in their infancy phase and pose more problems than opportunities (Zeidan & Itani, 2020).

There are other significant challenges the HR professional may encounter in embracing analytics. Despite the high interest in data analytics and their benefits, many challenges regarding using HR data hinder HR advancements. HR professionals may have to dive deep into their tools and techniques to unlock their full potential.

Adopting HR analytics requires analytically skilled personnel within the HR department instead of one person or a group of analysts familiar with all organization functions. Only organizations that create and maintain a balanced blend of relevant capabilities will succeed in HR analytics.

HR analytics' success will only come to those companies that can develop and maintain a balanced blend of pertinent competencies. Therefore, using data and big data to support strategic decision-making, enhance HR processes, and provide answers to workforce-related concerns is still seen as futuristic (Zeidan & Itani, 2020).

15.6 TRENDS AND TOOLS IN WORKFORCE ANALYTICS

Workforce analytics as a discipline for career transition was valued at USD 765.72 million in 2020. The growth expectation by 2026 is at a CAGR of 15.64%. Workforce analytics are used to study and learn information about people or the behavior of employees. This helps in making better workforce decisions by improving the productivity of existing employees and selection criteria, thus resulting in the retention of existing experienced employees.

HR analytics are the first step in exploring the various aspects of an organization's operations and business strategy. This method can yield a high ROI for the company due to its ability to transform how an organization operates (Delbridge & Barton, 2002).

Implementing HR analytics in management processes is often regarded as a critical component of a strategy, but it can also take time to answer questions about specific measures. For instance, when an organization has turnover, this can have a long-term impact on the business.

The organization's second necessary process is an evidence-based workforce analytics approach. This method helps the organization to get a better yield by using technological tools to predict the performance of the organization. The challenge is identifying and answering those key questions and determining which technology and tool be used (Marler & Boudreau, 2017).

Dulebohn and Johnson (2013) discussed the third trend as effective organizational performance; the organization should involve HR analytics as a decision-making tool. This trend has higher ROI because it improves the efficiency of the organization's decision-making.

15.6.1 HR Information System

A human resources information system (HRIS) is a database storing employee information. All personal data can be entered into the system, accessible from anywhere, 24/7. HRIS also often provides solutions for various sub-functions of HR, such as salary, benefits, learning, performance, recruitment, and more. Most are designed to be flexible integrated databases, with comprehensive functionality and reporting capabilities that enable better HR management. This technology aims to standardize the organization's HR rules and practices into a process that reflects an organization, and many popular technology companies provide such software.

An HRIS should enable more precise data gathering, increased HR staff productivity (for example, streamlining the resume collection process and automating time-off requests), quicker approvals, less paperwork, and better efficiency and quality in HR decision-making.

15.6.2 HR Data Warehouse

To put it simply, an HR data warehouse is a database that compiles and arranges historical talent data and other HR data for query requests and analysis. Data from various internal systems must be linked to make this system used.

To this purpose, it is crucial to regularly and periodically export data from all applicable systems of record in use and to store the data in a warehouse designed specifically for HR data. Staging areas are where data from operating systems are prepared, cleaned, and validated. The HR data warehouse receives such data, and a master record is created at the warehouse. This information can then be utilized to populate other downstream systems that require precise information for either advanced analytics or HR reporting needs (Khang et al., 2022a).

15.6.3 Reporting Technology

Users can do tasks like querying the HRIS for data on each HR sub-function, such as payroll reports, the typical time it takes to fill vacancies, the amount of engagement with online learning systems, etc., using reporting apps.

Business information tools can aid in identifying business issues that need to be resolved, which can support strategic and tactical decision-making processes. Additionally, business information technology can assess how well the HR function performs compared to a set of key performance indicator (KPI). Typically, this entails filling out dashboards with HR analytics, such as turnover rates across business units and roles (Khang et al., 2022c).

15.6.4 Statistics and Machine Learning

The organization problem will be treated as hypothesis and can be answered or solved using statistics and machine learning techniques. Many software and tools are available to run statistics and machine learning, such as Python, R, SPSS, Excel, etc.

Artificial intelligence may be used in machine learning technology to build up algorithms that could study past choices and produce internal designs that assist with data prediction. To create accurate predictions, it is ideal that these designs be based on historical trends and data relationships. For instance, machine learning models can predict expected future labor turnover specific to certain roles as an outcome of job dissatisfaction or other pertinent variables (Khanh & Khang, 2021).

15.6.5 Visualization

Visualization technology software can produce complex graphs and other visual representations of information. They help with data interpretation and aid analysts in enabling a more precise knowledge of the information they are considering, such as visual mapping of relationships between variables, thus aiding in comprehending the results better.

The potential benefit of this tool is that it is suitable even for individuals or managers who are not well-versed in statistics as a discipline. There are a few popular tools with straightforward graphical user interfaces, such as Tableau and PowerBI (Khang et al., 2023c).

15.6.6 Cognitive Computing

This is an emerging area of analytics where beyond the technological requirements essential for most analytical tools, the underlying algorithms and processes mimic human cognitive processes. This is an advancement to the existing visualization tools, where the outcomes of visual descriptions are combined with human cognitive abilities, which are replaced with system-driven reasoning and inferences (Rana et al., 2021).

Thus, unlike regular analytical tools, cognitive analytics use a probabilistic method to derive multiple answers to varying levels of relevance (Gudivada et al., 2016).

15.6.7 Robotic Process Automation

According to Khanh et al. (2022), robotic process automation (RPA) reduces the likelihood of errors and faults by automating work currently done manually. Software programs used in RPA tools and procedures run on virtual servers and can be started and stopped at any moment. Companies hire and train workers for problem-solving and brainstorming activities employing bots rather than repetitive robotic duties (Tailor et al., 2022).

15.6.8 Real-Time Analytics

Workforce analytics come with real-time data of the workforce, which helps the employee to get real-time feedback and learn from data for year-round improvement. The organization will be able to respond to the dynamic situation with improved competency and accuracy. It helps to measure the employee's productivity in real time and make decisions faster.

15.7 ANALYSIS OF DATA

This chapter analyzes and draws inferences from the workforce-related data of 1,470 employees of IBM. The HR data was made available at Kaggle.com as shown in Figure 15.1.

The visualization shows that the longer the duration served by an employee at the firm, the higher the chance is that he/she will continue with it. This is also related to job satisfaction, since employees with longer service have increased job satisfaction.

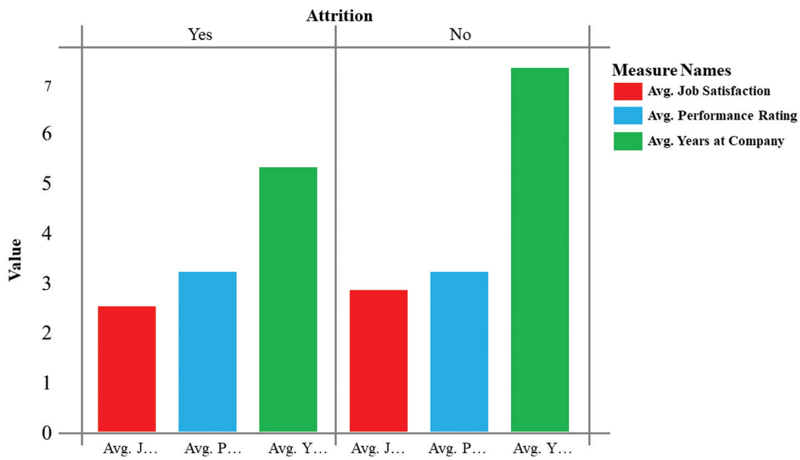

FIGURE 15.1 Relationship of attrition numbers.

However, job performance ratings do not show any significant difference between individuals leaving the firm and those willing to remain/continue with the firm, as shown in Figure 15.2.

The attrition seems to increase with an average age of 31, and it is more common with single individuals. In contrast, the minor attrition numbers can be seen with divorced individuals at an average age of 35, as shown in Figure 15.3.

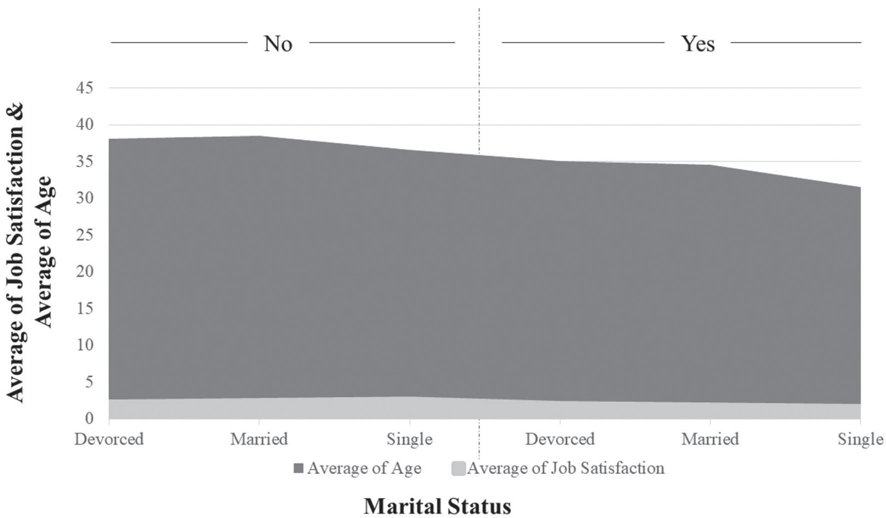

FIGURE 15.2 Relationship of attrition numbers based on job satisfaction, age, and marital status.